CSC 3114 Cryptology and Coding Theory

Revision Questions

Question 1

(a) Draw a labelled digital communication model showing the source, source encoder, channel encoder, modulator, noisy channel, demodulator, channel decoder, source decoder and sink in the correct order.

(b) A three-bit channel code uses the mapping

$$00 \mapsto 000, \quad 01 \mapsto 011, \quad 10 \mapsto 101, \quad 11 \mapsto 110.$$

A. List all words obtainable from 000 by exactly one bit error. Explain how the code detects each error.

B. If 100 is received and exactly one bit error occurred, list every possible transmitted codeword. Explain why correction is ambiguous.

C. Calculate the code rate and the number of redundant bits per message.

(c) A five-bit channel code uses the mapping

$$00 \mapsto 00000, \quad 01 \mapsto 01111, \quad 10 \mapsto 10110, \quad 11 \mapsto 11001.$$

I. Determine the minimum Hamming distance by showing the distance between every pair of distinct valid codewords.

II. The received word is 10000. Assuming at most one bit error, identify the transmitted codeword and recover the two-bit source word. Justify your answer using the distances from the received word.

III. Calculate the code rate.

(d) A two-bit source word is repeated five times; for example, $01 \mapsto 0101010101$.

i. Decode 1100100010 by applying majority voting separately to odd-numbered and even-numbered positions.

ii. State the number of bit errors this scheme is guaranteed to correct in each ten-bit codeword.

iii. Calculate its code rate.

Question 2

(a) Distinguish cryptography, cryptanalysis and cryptology.

(b) Define plaintext, ciphertext and a cryptographic key.

(c) Explain confidentiality and integrity. Give one example of a violation of each.

Question 3

(a) DES uses a Feistel network to transform plaintext blocks into ciphertext blocks.

i. State the effective key length of DES and explain why it is insufficient for modern security.

ii. In a simplified Feistel round, the input halves are

$$L_0 = 1010, \qquad R_0 = 1100,$$

and the round function produces $F(R_0, K_1) = 0111$. Calculate the output halves L_1 and R_1. Show your working.

iii. Using the Feistel round equations, explain why the original input halves can be recovered even when the round function F is not invertible.

(b) Describe the four transformations in a normal AES encryption round, in their correct order. State the purpose of each transformation.

(c) Consider the use of block cipher modes to encrypt messages containing several blocks.

i. Explain why ECB mode reveals patterns when a message contains repeated plaintext blocks.

ii. Two plaintext blocks P_1 and P_2 are encrypted in CTR mode using the same key and the same nonce–counter input, producing the same keystream block S:

$$C_1 = P_1 \oplus S, \qquad C_2 = P_2 \oplus S.$$

Derive an expression for $C_1 \oplus C_2$ and explain what it reveals.

iii. If an attacker knows P_1, show how the attacker can recover P_2. State the rule for CTR inputs that prevents this keystream reuse.

(d) A financial system transmits the message $M =$ "Transfer UGX 100,000" together with its unkeyed hash $h = \mathrm{Hash}(M)$. Both are sent over an untrusted network.

I. Explain how an active attacker can alter the message and still make the receiver's hash comparison succeed.

II. Explain how replacing the unkeyed hash with an HMAC addresses this weakness.

Question 4

(a) Draw a labelled encryption–decryption model for Alice sending a message to Bob over an insecure channel. Include the key, plaintext, ciphertext, and Eve.

(b) Distinguish symmetric encryption from asymmetric encryption in terms of the keys used and which keys must remain secret.

(c) Eve modifies Alice's message and sends it to Bob while pretending to be Alice. Identify two security goals threatened.

Question 5

(a) Distinguish symmetric encryption from asymmetric encryption in terms of the keys used for the following operations. For each operation, identify any keys that must remain secret.

i Encryption.

ii Decryption.

(b) Distinguish between the following in terms of their main purposes.

i. Coding theory and cryptography.

ii. Source coding and channel coding.

(c) Explain each of the following approaches to error control, stating what the receiver does after an error is detected.

i. Backward error correction (BEC).

ii. Forward error correction (FEC).

(d) Distinguish steganography from digital watermarking in terms of:

A. their primary purposes;

B. the relationship between the hidden information and its carrier.

(f) Define the following terms.

i. Cover object.

ii. Payload.

iii. Stego-key.

iv. Stego-object.

(g) Explain the following concepts in block cipher design. For each concept, identify a transformation that contributes to it.

i. Confusion.

ii. Diffusion.

(h) Consider block ciphers and stream ciphers.

i. Distinguish them in terms of how plaintext is processed.

ii. Give one example of each.

(i) Explain each of the following concepts and illustrate each using a university student portal.

i. Authentication.

ii. Authorisation.

(j) Explain each of the following authentication approaches, giving an example of each.

I. One-way authentication.

II. Mutual authentication.

Question 6

(a) A client authenticates to a server by sending its identity A and the same token $T = \mathrm{MAC}_K(A)$ on every login. The client and server share a secret key K.

i. Explain how an attacker can impersonate the client without knowing K.

ii. Using a labelled message sequence, describe a one-way challenge–response protocol that addresses this weakness. Specify the information authenticated by the response and the checks the server must perform before accepting the client.

(b) The university wishes to protect IP traffic between two campus gateways over a public network.

i. Select a suitable IPsec packet-protection protocol and mode. Justify your choices.

ii. State the role of IKEv2 in establishing this protection.

iii. Explain why this arrangement does not automatically protect traffic between a campus host and its local gateway.

(c) The student portal uses a certificate-based TLS 1.3 handshake with ephemeral Diffie–Hellman key establishment. Let

$$p = 23, \qquad g = 5, \qquad x = 4, \qquad y = 3,$$

where x and y are the client's and server's private values, respectively.

A. Calculate the public values $X = g^x \bmod p$ and $Y = g^y \bmod p$.

B. Calculate the shared secret independently at the client and server. Show your working.

C. Explain the distinct purposes of `CertificateVerify` and `Finished` in the TLS 1.3 handshake.

(d) A staff member must email a confidential attachment through university mail servers. The recipient must also verify the sender's identity and detect changes to the attachment.

i. Explain why TLS protection on every connection between mail clients and servers does not, by itself, keep the attachment confidential from those servers.

ii. Describe how message encryption and a digital signature can meet these requirements using S/MIME or OpenPGP. Identify whose keys are used for each operation.

Question 7

(a) Distinguish between steganography and cryptography in terms of:

i. their primary goals;

ii. the outputs produced.

(b) Using a labelled diagram, describe the basic steganographic embedding and extraction model. Your diagram should show the cover object, payload, stego-key, embedding algorithm, stego-object and extraction algorithm.

(c) A sender intends to hide confidential information in a 640×480 RGB image using one least significant bit (LSB) from each of the three colour channels.

i. Calculate the raw embedding capacity in bits, bytes and KiB. Take $1\,\text{KiB} = 1024$ bytes. Show your working.

ii. The confidential information occupies 100 KiB, while headers, encryption metadata and authentication tags require an additional 8 KiB. Determine whether the image has sufficient capacity. Justify your answer.

iii. After embedding, the image is converted to JPEG. Explain why the hidden information may not be recovered correctly.

(d) Explain the security benefit of each of the following:

i. encrypting the payload before embedding it;

ii. using a secret stego-key when the embedding algorithm is publicly known.

Question 8

(a) Consider the Caesar cipher.

i. Encrypt `ATTACK AT ONCE` using a Caesar shift of 4.

ii. Decrypt `VHFUHW PHVVDJH` using a Caesar shift of 3.

(b) Encrypt `DEFEND THE EAST WALL OF THE CASTLE` using regular columnar transposition with the keyword `GERMAN`. Write the letters row by row, pad with `X` to complete the rectangle, and read columns from top to bottom in alphabetical order of the keyword letters.

i. Show the completed rectangle and the resulting ciphertext.

ii. Distinguish regular from irregular columnar transposition.

(c) Use the Vigenère cipher with the keyword `LUCKY`.

i. Encrypt `COMPUTING GIVES INSIGHT`. Show the repeated keyword and letter-by-letter working.

ii. Decrypt the first five ciphertext letters obtained in part (i), showing your working.

(d) State the four requirements for a secure one-time pad.